

Speaker Notes

QFT, Phase Estimation, and Shor's Algorithm

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1. Title slide

The punchline of this talk: quantum computers can factor big numbers fast, and the engine that makes it work is the Fourier transform — the same one from this course, applied to a quantum state. Four steps: DFT, QFT, phase estimation, Shor's. Don't memorize matrices; track what each piece is doing.

2. Roadmap

Each arrow means “the previous thing, used as a tool.” The QFT is the DFT in disguise. QPE is a clever use of the QFT. Shor's is QPE pointed at one special matrix. If the audience remembers one thing, it's this chain — the rest of the talk just fills in the arrows. Classically the best factoring algorithms are sub-exponential; Shor's is polynomial in the number of digits, which is why it breaks RSA in principle.

3. The DFT: a change of basis from roots of unity

Remind them: the DFT takes a list of numbers to a new list; each output is a weighted sum where the weights are roots of unity — points evenly spaced on the unit circle. The block is the one proof worth showing in full, and it's pure complex analysis: if N doesn't divide m , the powers of ω walk around the circle and the geometric series sums to zero, because $\omega^N = 1$ exactly. All the roots cancel. This cancellation is the reason the Fourier basis is orthonormal, hence the reason the DFT matrix is unitary, hence — much later — the reason the quantum circuit is even allowed to exist. One slide of complex analysis powers the whole talk.

4. Qubits (covered in class)

Skip slide — say “we did this in class.” A qubit is a unit vector in $\mathbb{C}^2$; n qubits live in a 2^n -dimensional space via tensor products. The picture is the Bloch sphere: after modding out the global phase, a single qubit state is a point on an ordinary sphere — and for us complex analysts, that sphere is the Riemann sphere: the ratio β/α lives in $\mathbb{C} \cup \infty$, and stereographic projection

puts it on S2. North pole is 0, south pole is 1. Gates act as rotations of this sphere. Move on quickly.

5. The gate set

Four gates, the whole toolkit. Hadamard turns a definite 0 or 1 into an even mix — it creates superpositions. R-k leaves 0 alone and spins the 1-component by a fixed angle: it writes a phase. SWAP exchanges two wires. Controlled R-k fires the phase only when the control qubit is 1 — point at the bottom-right entry, that single $e^{2\pi i/2^k}$ is its only action. On the Bloch sphere picture from the previous slide, the R-k gates are rotations about the vertical axis. Everything in the QFT circuit is built from just these.

6. QFT: the DFT factorizes over qubits

The QFT is the DFT applied to amplitudes of a quantum state. Naively that's a 2^n by 2^n matrix — hopeless. But when N is a power of two, the output miraculously splits into a tensor product of single-qubit states; that's the formula, with $0.j\dots$ meaning a binary fraction. Each factor is one qubit holding one phase, built by a Hadamard plus a few controlled rotations. So an exponential-size transform collapses to about n-squared gates. Plain-language version to say out loud: the QFT takes information stored in amplitudes and re-writes it as phases — angles — spread across the qubits. Hold onto that; phases are exactly what the next tool reads out.

7. QFT circuit

Walk it left to right: each wire gets a Hadamard, then controlled rotations — the S, T, R4, R5 boxes are R-k gates — reach over from the lower qubits to stamp in smaller and smaller phase corrections, the binary-fraction digits from the last formula. The crossings at the far right are SWAPs putting the qubits back in order. Don't trace every gate; the point is the circuit is short and regular — that regularity is the n-squared efficiency made visible.

8. QPE: eigenvalues of unitaries are phases

Second tool. A unitary preserves norms, so the one-line proof on the slide shows every eigenvalue has modulus one — eigenvalues live ON the unit circle, as in the picture. So an eigenvalue is nothing but an angle theta. Phase estimation asks: given the matrix and an eigenvector, find that angle. The catch: applying U to its eigenvector only multiplies by a global phase, and global phases are invisible to measurement — probabilities only see absolute values. So we need a trick to move theta somewhere measurable. That trick is phase kickback, next slide.

9. Phase kickback

The cleverest move in the talk; slow down here. Normally the control decides and the target gets acted on. We flip it: put the CONTROL in superposition and feed the TARGET an eigenvector. On the 0-branch nothing happens; on the 1-branch U hits ψ — but U on its eigenvector is just multiplication by $e^{2\pi i\theta}$, a number, which slides off ψ and attaches to the 1-branch of the control. End result: ψ is untouched, and the phase has jumped “backwards” onto the control qubit. The gate ran in reverse, in a sense — the target reprogrammed the control. That rescues the invisible phase: it now sits on a qubit we can play with.

10. One ancilla: estimate theta by sampling

This circuit is the smallest possible phase estimator: Hadamard, controlled- U , Hadamard, measure. The second Hadamard makes the 0 and 1 branches interfere, and the interference angle is θ — giving the biased coin on the slide: probability $\cos^2$ of $\pi\theta$ for 0, $\sin^2$ for 1. So run it a thousand times, count zeros, solve backwards for θ — a Monte Carlo estimate, like estimating a coin’s bias by flipping it repeatedly. It works but it’s slow: lots of shots for a few digits. Next slide: get all the digits in ONE shot.

11. One ancilla: where cos-squared comes from

This is the missing algebra behind the coin-flip from the previous slide. The phase kickback left the ancilla as an even mix of 0 and 1 with a phase $e^{-2\pi i\theta}$ hanging on the 1. Apply Hadamard once more, using its action on the two basis states at the top. The 0-components add, the 1-components subtract, so the amplitudes become one-half of one-plus-the-phase on 0, and one-half of one-minus-the-phase on 1. Then the Born rule: probability is amplitude-magnitude-squared. Expand using the identity that the modulus squared of one-plus- e^{-ix} is two-plus-two-cosine- x — a half-angle collapse turns 0’s probability into cosine-squared of $\pi\theta$ and 1’s into sine-squared. Sanity checks to say out loud: if θ is 0, the phase is 1, the 1-branch cancels completely, and you measure 0 with certainty — makes sense, no phase to detect. If θ is one-half, you get 1 with certainty. In between, the bias of the coin tracks θ , which is exactly what lets the Monte Carlo estimate work.

12. Many ancillas: the kickback state is a QFT

Use t ancillas; the k -th one controls U to the power 2^k , so it kicks back θ doubled k times — each ancilla captures a different binary digit of θ . Now compare the resulting product state with the QFT factorization from earlier: identical. The pile of kicked-back phases IS the Fourier transform of θ written in binary. θ is hiding in Fourier space — and we own the machine that inverts that. Run the inverse QFT, measure, and the binary digits of θ drop out directly. One shot, no statistics; more ancillas means more digits. THIS is why we built the QFT first: it’s the read-out device for phases.

13. QPE circuit

The whole algorithm in one picture, three blocks: Hadamards make the superposition; the controlled powers of U do the kickback, loading θ into Fourier space; the inverse-QFT box decodes it and we measure the binary answer. Bottom wire is the eigenvector, which just sits there donating its phase — note it comes out unchanged.

14. Shor's: the number theory

The classical half. Pick a coprime to N ; its powers cycle with period r , the order. If r is even, set $b = a$ to the r -over-2, so b squared is 1 mod N , and b -squared-minus-1 factors as $(b-1)(b+1)$. N divides that product. Now the key step — say it slowly: provided b isn't congruent to plus or minus 1, N divides NEITHER factor alone. The only way to divide a product but neither piece is for N 's primes to SPLIT — some land in $(b-1)$, the rest in $(b+1)$. So $\gcd(b-1, N)$ scoops up exactly the primes that landed on the left: it can't be 1 (some are there) and can't be N (not all are), so it's an honest factor. And gcds are cheap — Euclid's algorithm. Random a works with probability at least one half. Conclusion: factoring reduces to finding r .

15. Shor's: order-finding is QPE

The quantum half: choose U to be multiplication by a , mod N . It's unitary because multiplication by an invertible element just permutes the basis states. And because applying it r times returns you to start, its eigenvalues are exactly the r -th roots of unity, e to the $2\pi i s$ over r — the same circle points from slide one. So QPE reads off a phase s over r , and the denominator hands us the order. Pipeline: measure a decimal close to s/r , clean it into an exact fraction with continued fractions, read r , finish with two gcds. Worked example: $N = 15$, $a = 7$. The order is 4 since 7-to-the-4th is 2401, which is 1 mod 15. Then $b = 49 = 4$ mod 15, and $\gcd(3, 15) = 3$, $\gcd(5, 15) = 5$: out pops $15 = 3$ times 5. One detail I'm sweeping under the rug: we can't prepare an eigenvector since they depend on r — but the state “1” happens to be an equal superposition of all of them, and QPE handles the superposition by linearity, returning a random s over r . That's in the paper if anyone asks.

16. Takeaways

Close the loop by running the chain backwards: Shor's needs the order; order-finding is phase estimation; phase estimation reads phases with the inverse QFT; and the QFT exists because roots of unity are orthogonal — the complex analysis we started with. Honest caveat to end on: today's hardware doesn't have nearly enough error-corrected logical qubits to factor anything RSA-sized, so this is a theorem about the future. Stop and take questions.